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MECH 5800 - Advanced Optimization Modelling

A Cost-Benefit Framework for Bus Allocation Optimization

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Abstract

Public transit systems face mounting pressure to deliver equitable, efficient service under tight fiscal constraints. This report develops a cost-benefit framework to formulate a mixed-integer linear programming (MILP) model for optimizing bus allocation across OC Transpo routes in Ottawa, Ontario, with the objective of maximizing socio-economic benefits within a constrained operating budget. The model quantifies benefits and costs associated with changes in service frequency, incorporating non-market valuation techniques to monetize passenger waiting time savings and greenhouse gas emission reductions resulting from mode shift away from personal vehicles. Operating costs, including driver wages, fuel, and maintenance, are estimated from publicly available Canadian data sources.

Ridership response to frequency changes is modelled using a demand elasticity of 0.5, and nonlinear benefit terms are linearized via first-order Taylor expansion to enable integration with standard MILP solvers. The model is implemented in Python using the Gurobi optimizer and applied to 2023 weekday ridership data and GTFS schedule data for OC Transpo routes. The optimized bus allocation yields an estimated 20% increase in weekday ridership with a net cost of approximately $12,700 against an allowable $50,000 increase in operation and maintenance costs.

Sensitivity analysis reveals that the solution is primarily governed by fleet capacity constraints and limits on per-route bus changes, rather than by budget or parameter uncertainty. Cost and benefit parameters are shown to have negligible influence on the optimal allocation across the range tested. Future work should address the limitations of the linearization in capturing diminishing returns of increased demand, incorporate more recent and higher-resolution ridership data, and account for differences in capacity across bus types within the fleet.

1 Problem Introduction

Public transport is an essential part of any metropolitan city for both social and logistical reasons. Primarily transit operates to provide "a low-cost cross town means of travel for people who cannot or will not drive" [1]. From a logistical perspective, in a large urban area, a transportation system based solely on personal vehicles would be unable to cope with the density and volume of travel required to keep a city functioning efficiently [2]. Despite the essential role of public transit, cities are often constrained by tight budgets, high operating costs, and decreased ridership following the COVID-19 pandemic.

From this perspective, there is a clear motivation to improve service quality without increasing overall costs. Accordingly, this project is driven by the objective of formulating an optimization framework that improves transit system performance based on social considerations, without requiring capital investment or significant increases in operating and maintenance expenditures.

1.1 OC Transpo

The selected transport system for this project is OC Transpo, which is the public transit service for the city of Ottawa, Ontario, operating both a light rail system (O-Train) and a bus network. The bus network includes 27 *Frequent* routes, which run at intervals of 15 minutes or less, *Connexion* routes, designed to connect residential communities to the O-Train in the morning and return trips in the afternoon, and *Local* routes, which serve a variety of areas. In total, the system serves approximately 300,000 daily riders across both modes and maintains a fleet of over 700 vehicles [3].

OC Transpo reported a budget deficit of $52 million [4] in 2025 and faces increasing financial constraints associated with O-Train expansions. However, the bus system remains essential for providing mobility across Ottawa. Accordingly, the goal is to reallocate the existing bus fleet to maximize the social benefits of the transport system while maintaining the current route network in Ottawa.

1.2 Project Objectives

The objectives of this project are as follows:

1. Evaluate public transit benefits using non-market valuation techniques
2. Develop a cost-benefit analysis to quantify the impacts of changes in service frequency
3. Optimize route frequency across selected OC Transpo bus routes to maximize social benefits within a constrained operating budget
4. Conduct sensitivity analysis to address uncertainty and identify key parameters influencing results

2 Methods

2.1 Model Formulation

The optimization of select OC Transpo bus routes is based on an economic framework, specifically a cost-benefit analysis that incorporates non-market valuation methods to quantify the social benefits associated with changes in bus route frequency. The optimal system is defined as one that allocates the available bus fleet across routes in a manner that maximizes overall socio-economic benefits. This leads to an optimization problem in which an integer number of buses must be assigned to each route for a given time period of the day.

The current OC Transpo system serves as the baseline against which changes in service frequency are evaluated. To characterize the current system passenger demand data and General Transit Feed Specification (GTFS) data is utilized. The ridership data aggregates the number of passengers on a given route over a time block during the day, as described in Table 1, which limits the temporal resolution of the model to an optimal number of buses during each time block. Consequently, route parameters like service frequency and trip length will be constant and averaged over a time block.

Table 1: Time blocks

Time Block	Time Period
Early AM	3:00-6:30
AM Peak	6:30-9:00
Midday	9:00-15:00
PM Peak	15:00-18:30
Evening	18:30-23:00
Night	23:00-3:00

The model formulation results in a mixed-integer programming (MIP) problem, where the decision variable is the number of buses assigned to each route. Any nonlinearities are addressed through first-order linearization techniques.

2.2 Evaluating Public Transport Benefits and Costs

There is a substantial body of academic literature on methods for evaluating the costs and benefits of non-market impacts in project analysis. These approaches are particularly important for public sector projects, where many benefits related to the public good such as public health, and the environment are not directly reflected in market prices. Public transportation is a clear example of a service that generates both internal and external benefits. Improvements to transit service enhance convenience and comfort for passengers, while also producing broader societal benefits such as reduced road congestion, lower air pollution, and decreased greenhouse gas (GHG) emissions.

Although the benefits and costs associated with transit projects are wide-ranging, the scope of this analysis is limited to three primary effects: the impact of increased bus frequency on ridership, the benefits of reduced passenger waiting times, and the environmental benefits arising from decreased personal vehicle use. A summary of the in scope benefits and costs is provided in Table 2. The economic modelling approach adopted in this analysis is primarily informed by the methodologies presented in [2], [5], and [6].

Table 2: In scope benefits and costs for increased transit service frequency

Improvement	Benefits	Costs
Increased service frequency	- Reduced passenger wait times - Reduced GHG emissions per km travelled	- Increased maintenance costs - Increased fuel costs - Increased number of bus drivers

The benefits in Table 2 require non-market valuation of travel time savings and avoided emissions, and therefore involve inherent uncertainty and subjectivity. In contrast, the in-scope costs have observable market values (maintenance, fuel, and bus driver wages) and are inherently quantitative. However, estimating representative average values for these costs still involves some uncertainty. As a result, sensitivity analysis will be conducted on all estimated parameters to evaluate the robustness of the results.

2.3 Cost and Benefit Equations

Before formulating the global optimization problem (i.e., the minimization of costs and maximization of benefits) the equations used to quantify the relative change in costs and benefits from the baseline system are first introduced. The variables $n_{new,r,t}$ and $n_{old,r,t}$ represent the number of buses in the new system and current system, respectively. The indices r and t denote the given route and time block, respectively. All parameters and their values are summarized in Section 2.6 in Table 3.

2.3.1 Labour Cost

The change in labour cost for a change in the number of buses for route r and time block t is calculated as,

$$\Delta C_{labour,r,t} = (n_{new,r,t} - n_{old,r,t}) \cdot H_{block,t} \cdot W_{driver} \tag{1}$$

where $H_{block,t}$ is the number of service hours within the given time block, t, and W_{driver} is the hourly wage rate of an OC Transpo bus driver.

The hourly wage rate was taken directly from the City of Ottawa bus operator rates of pay [7] and is calculated as an equal weighted average of the four tiers of pay which range from $31.28-$36.80/hr.

2.3.2 Fuel Cost

The change in fuel cost for a change in the number of buses for route r and time block t is calculated as,

$$\Delta C_{fuel,r,t} = (n_{new,r,t} - n_{old,r,t}) \cdot \overline{L}_r \cdot \frac{H_{block,t}}{\overline{T}_{r,t}} \cdot P_{fuel} \cdot FC_t \tag{2}$$

where $\overline{L}_r$ is the average route distance, $\overline{T}_{r,t}$ is the average time duration of the route, P_{fuel} is the price of fuel per litre, and FC_t is the fuel consumption rate of a bus in litres per km travelled.

The distance of the route $\overline{L}_r$ was calculated from GTFS data and was calculated as an average value across all time blocks. The time duration of the route $\overline{T}_{r,t}$ was also calculated from GTFS data and was calculated as an average for each time block. The price of fuel P_{fuel} was taken from Statistics Canada [8] and was calculated as an average over 6 months in 2025 for Ottawa. The fuel consumption FC_t was taken from National Resource Canada [9] and is representative of an average fuel consumption rate for transit buses in 2023. During peak travel time blocks (AM Peak and PM Peak) FC_t is taken as 10% higher than the average, and in off peak travel time blocks (Early AM and Night) FC_t is taken as 10% lower than the average. This accounts for higher fuel efficiency in low-traffic versus congested conditions.

2.3.3 Maintenance Cost

The change in maintenance cost for a change in the number of buses for route r and time block time t is calculated as,

$$\Delta C_{maintenance,r,t} = (n_{new,r,t} - n_{old,r,t}) \cdot \overline{L}_r \cdot \frac{H_{block,t}}{\overline{T}_{r,t}} \cdot P_{maintenance} \tag{3}$$

where $P_{maintenance}$ is the cost of maintenance per km travelled.

The price of maintenance per km travelled $P_{maintenance}$ is taken from Statistics Canada [10] and increased slightly to account for the fact the latest figure is from 2013.

2.3.4 Elasticity of Demand

Transit service quality, measured through factors such as speed, comfort, reliability and safety, is shown to have an affect on ridership [5, 6]. Discretionary travellers (those who have access to a private vehicle but choose to use transit) are particularly sensitive to service quality. Higher service quality reduces perceived travel time costs, making transit a more attractive option for this group [5], and can induce a mode shift from car travel to transit for a given trip.

One key component of service quality is service frequency, which directly affects user convenience and waiting times. As a result, increases in service frequency are generally associated with higher ridership. According to [6], the average elasticity of transit demand with respect to service frequency is approximately 0.5. This implies that a 1% increase in service frequency leads to an estimated 0.5% increase in ridership.

The result of this is that given a current known ridership demand $x_{old,r,t}$, the new ridership $x_{new,r,t}$ can be estimated as,

$$x_{new,r,t}(f) = x_{old,r,t} \left(\frac{f_{new,r,t}}{f_{old,r,t}} \right)^{0.5} \tag{4}$$

where $f_{new,r,t}$ and $f_{old,r,t}$ is the service frequency for the given route in the new and current system, respectively.

Instead of expressing service frequency as a continuous variable, the demand elasticity can be reformulated in terms of the integer number of buses assigned to a route. Accordingly, the updated ridership demand can be expressed as follows,

$$x_{new,r,t}(n) = x_{old,r,t} \left(\frac{\frac{n_{new,r,t}}{\overline{T}_{r,t}}}{\frac{n_{old,r,t}}{\overline{T}_{r,t}}} \right)^{0.5} \tag{5}$$

$$= x_{old,r,t} \left(\frac{n_{new,r,t}}{n_{old,r,t}} \right)^{0.5} \tag{6}$$

2.3.5 Value of Time Benefit

To monetize travel time, unit costs are typically estimated relative to average wages, with adjustments based on several contextual factors [5]. These factors include whether the travel time is personal (unpaid) or work-related (paid), as well as qualitative aspects such as convenience, comfort, and security [5]. For example, individuals generally perceive the cost of travel during peak periods or while standing on a crowded bus as significantly higher than travel under more comfortable and less congested conditions.

Of particular relevance to our analysis is the value of waiting time. According to [11], the unit cost of waiting is typically estimated at two to five times higher than in-vehicle transit travel time, while [12] recommends using 100% of the wage rate. Additionally, travel time costs are generally lower for children and for individuals who are retired or unemployed compared to those with full-time employment [5].

In our model, the value of travel time is set at 70% of the hourly wage during off-peak periods and 100% of the wage rate during peak AM and PM periods. These values were chosen to reflect differences in travel conditions and passenger composition. Peak periods are typically associated with slower, more congested traffic and a higher proportion of full-time workers, while off-peak periods generally offer more comfortable travel conditions and a more diverse mix of passengers, including university students, unemployed individuals, and leisure travellers.

Therefore, the benefit associated from the reduced waiting time for passengers due to a change in the number

of buses for route r and time block time t is calculated as,

$$\Delta B_{time,r,t} = \frac{1}{2}\left(\frac{1}{f_{old,r,t}} - \frac{1}{f_{new,r,t}}\right) \cdot x_{new,r,t} \cdot W_{passenger} \cdot F_{wage,t} \tag{7}$$

$$= \frac{\overline{T}_{r,t}}{2}\left(\frac{1}{n_{old,r,t}} - \frac{1}{n_{new,r,t}}\right) \cdot x_{old,r,t}\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5} \cdot W_{passenger} \cdot F_{wage,t} \tag{8}$$

where $W_{passenger}$ is the average hourly wage rate and $F_{wage,t}$ is the fraction of the hourly wage rate used to value the travel time saved of each passenger. The average waiting time of a passenger is estimated as half of the headway time.

2.3.6 Greenhouse Gas Emissions Reduction Benefit

An increase in service frequency has been shown to reduce car travel [6], which consequently lowers greenhouse gas emissions. In general, transit-oriented communities tend to have fewer personal vehicles and a lower number of vehicle miles travelled per capita [6]. Newman and Kenworthy found that, across 23 cities in the United States, Canada, Australia, and Europe, each transit passenger-mile corresponds to a reduction of 3.6 automobile vehicle-miles (5.8 km) [13]. This effect can be explained by the fact that car travel is generally faster than public transit, leading individuals to drive longer distances and travel more frequently than those using transit. Additionally, the emission intensity of transit is generally lower than that of private vehicles, except during periods of low passenger volume [14].

The benefit of avoided vehicle emissions for an increase in service frequency for route r and time block t is calculated as,

$$\Delta B_{emissions,r,t} = I_{GHG,car,t} \cdot V_{km,saved} \cdot (x_{new,r,t} - x_{old,r,t}) \cdot \overline{L}_{r,passenger} \cdot SCC \tag{9}$$

$$= I_{GHG,car,t} \cdot V_{km,saved} \cdot \left(x_{old,r,t}\left(\frac{n_{old,r,t}}{n_{new,r,t}}\right)^{0.5} - x_{old,r,t}\right) \cdot \overline{L}_{r,passenger} \cdot SCC \tag{10}$$

where $I_{GHG,car,t}$ is the GHG emission intensity per passenger km travelled in a car, $V_{km,saved}$ is the vehicle-kilometres saved per 1 km of transit passenger km travelled, $\overline{L}_{r,passenger}$ is the average length of the passenger's commute and SCC is the social cost of carbon.

The emission intensity for personal vehicles was obtained from [14] and is shown to vary throughout the day, with the highest intensities occurring during peak hours due to congested traffic. To approximate the distance travelled by a passenger, it was assumed to be 20% of the total bus route length, $\overline{L}_r$; that is, $\overline{L}_{r,passenger} = 0.2\overline{L}_r$. Finally, the social cost of carbon was taken from values used by the Government of Canada for emissions costing activities. Specifically, the 2026 value with a 2% near-term Ramsey discount rate was applied.

2.3.7 Linearization

For the ease of application of a mixed-integer linear programming (MILP) solver the non-linear terms introduced in the benefit equations were linearized. The selected method to linearize the equations is a first order Taylor expansion.

The first order Taylor expression around a point x_0 is defined as:

$$f(x) = f(x_0) + f'(x_0) \cdot (x - x_0) \tag{11}$$

In Equation 8 the non-linear term is:

$$\left(\frac{1}{n_{old,r,t}} - \frac{1}{n_{new,r,t}}\right)\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5}$$

The linear approximation is:

$$\left(\frac{1}{n_{old,r,t}} - \frac{1}{n_{new,r,t}}\right)\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5} \approx \frac{1}{n_{old,r,t}^2}\left(n_{new,r,t} - n_{old,r,t}\right) \tag{12}$$

Therefore the value of time saved equation can be expressed as:

$$\Delta B_{time,r,t} \approx \frac{\overline{T}_{r,t}}{2} \cdot W_{passenger} \cdot F_{wage,t} \cdot \beta \cdot (n_{new,r,t} - n_{old,r,t}) \tag{13}$$

where $\beta = \frac{x_{old,r,t}}{n_{old,r,t}^2}$.

In Equation 10 the non-linear terms is:

$$\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5}$$

The linear approximation is:

$$\left(\frac{n_{new,r,t}}{n_{old,r,t}}\right)^{0.5} \approx 1 + \frac{0.5}{n_{old,r,t}}(n_{new,r,t} - n_{old,r,t}) \tag{14}$$

Therefore the emissions cost benefit can be expressed as:

$$\Delta B_{emission,r,t} \approx I_{GHG,car,t} \cdot V_{km,saved} \cdot \overline{L}_{r,passenger} \cdot SCC \cdot \alpha \cdot (n_{new,r,t} - n_{old,r,t}) \tag{15}$$

where $\alpha = \frac{0.5 x_{old,r,t}}{n_{old,r,t}}$.

Thus, the system can be solved as MILP, which enables the use of a commercial MILP solver such as Gurobi.

2.3.8 Constraints

The model is subject to the following constraints.

The number of buses assigned across all routes for a given time block is constrained by a fleet total for that time block:

$$\sum_{r \in R} n_{new,r,t} \leq B_{total} \qquad \forall t \in T \tag{16}$$

For the standard configuration, this is taken as a 10% increase of the original value as estimated based on the 2023 ridership data presented in Section 2.5.1.

An operational budget constraint is proposed to restrict the costs associated with changes in the bus allocation:

$$\sum_{r \in R} \sum_{t \in T} (\Delta C_{labour,r,t} + \Delta C_{fuel,r,t} + \Delta C_{maintenance,r,t}) \leq \text{Budget}_{total} \tag{17}$$

In the simulation, $Budget_{total}$ was developed iteratively and settled at $50,000. Since the objective of this project is to study how bus allocations change under a constrained but nonzero budget, the parameter needed to be large enough to permit meaningful reallocation, while still remaining small relative to the scale of a municipal transit agency [4].

The limit on the change in buses from the baseline system is:

$$-\Delta n_{max} \leq n_{new,r,t} - n_{old,r,t} \leq \Delta n_{max} \qquad \forall r \in R,\ \forall t \in T \tag{18}$$

The standard-configuration value of $\Delta n_{\max} = 4$ was selected as a practical upper bound on route-level fleet reallocation within a single time block. The purpose of this constraint is to prevent the optimizer from making unrealistically large changes to service on individual routes, which would reduce the operational plausibility of the solution.

Finally, the decision variable must remain a non-negative integer:

$$n_{new,r,t} \in \mathbb{Z}_{\geq 0} \qquad \forall r \in R,\ \forall t \in T \tag{19}$$

2.4 Global Minimization Problem

$$\min_{n_{\text{new},r,t}} \quad \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}} \Delta\Big(C_{\text{labour},r,t} + C_{\text{maintenance},r,t} + C_{\text{fuel},r,t} - B_{\text{time},r,t} - B_{\text{emissions},r,t}\Big) \tag{20}$$

$$\text{s.t.} \quad \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}} \Big(C_{\text{labour},r,t} + C_{\text{maintenance},r,t} + C_{\text{fuel},r,t}\Big) \leq \text{Budget}_{\text{total}} \tag{21}$$

$$\sum_{r\in\mathcal{R}} n_{\text{new},r,t} \leq B_{\text{total},t}, \quad \forall t \in \mathcal{T} \tag{22}$$

$$\Delta n_{r,t}^{\min} \leq n_{\text{new},r,t} - n_{\text{old},r,t} \leq \Delta n_{r,t}^{\max}, \quad \forall r \in \mathcal{R},\ \forall t \in \mathcal{T} \tag{23}$$

$$n_{\text{new},r,t} \in \mathbb{Z}_{\geq 0}, \quad \forall r \in \mathcal{R},\ \forall t \in \mathcal{T} \tag{24}$$

$$B_{\text{total},t} \in \mathbb{Z}_{\geq 0}, \quad \forall t \in \mathcal{T} \tag{25}$$

$$\Delta n_{r,t}^{\min},\ \Delta n_{r,t}^{\max} \in \mathbb{Z}, \quad \forall r \in \mathcal{R},\ \forall t \in \mathcal{T} \tag{26}$$

For the minimum number of buses we assume that the frequency must be at least once per hour in off peak times and twice per hour for on peak times.

2.5 Data Sources and Parameter Estimation

The two primary data sources used in this analysis were ridership data for a sample month in 2023 and current route data published by OC Transpo. On April 27, 2025, OC Transpo launched *New Ways to Bus*, a significant overhaul of the bus network [15]. This initiative involved retiring some routes due to low ridership and improving connections to new routes and the O-Train. As a result of these updates, there is some mismatch between the ridership data used and the route data. Consequently, the model only optimizes bus routes for which 2023 ridership data are available. It is also acknowledged that these ridership trends may not fully reflect current patterns following the *New Ways to Bus* updates. Additional comments about the nature of each data source are as follows.

2.5.1 Ridership data

An excerpt of the ridership data is included in Appendix A.1. It features raw ridership numbers from January 2023 (among other months of 2023), sectioned by the time blocks defined in Table 1. It offers ridership both for weekday averages and weekend averages. For this project, the scope was narrowed to weekdays during January. It also counts the average daily ridership at approximately 200,000 which offers a baseline for comparison [16].

2.5.2 Route data

The data used to characterize the current system ($\overline{L}_r$, $\overline{T}_{r,t}$, $f_{old,r,t}$, $n_{old,r,t}$) was obtained from the GTFS data provided by OC Transpo. GTFS is a standardized data format that public transit agencies publish, containing timing and geographic information on routes and stops. It includes both static data, which describes predetermined scheduled information, and real-time extension data, which provides live transit updates such as the current location of a bus for trip planning platforms. In this case, the static data is appropriate.

As the ridership data aggregates the total number of passengers during a time block for both outbound and inbound routes, all route data is taken as an average of the two. Consequently, the average route length represents the mean of the outbound and inbound lengths during the given time block. While this is not always strictly accurate, it is a reasonable simplification for our purposes. The same principle applies to the average headway or frequency of a bus. At certain times of day, inbound service may be more frequent than outbound service; however, this is simplified to an average of the two due to the nature of the ridership data.

2.6 Parameters

The following table summarizes the parameters introduced in the benefit and cost equations. Additionally, the values and sources for constant parameters are indicated. In the Value column *Variable* indicates that the parameter depends on the route and is therefore not included in the table due to the high volume of values.

Table 3: Parameters summary

Parameter	Description	Value	Unit	Source
$n_{new,r,t}$	Number of buses in new system	Variable	-	Model
$n_{old,r,t}$	Number of buses in current system	Variable	-	GTFS [17]
$f_{new,r,t}$	Frequency of service in new system	Variable	1/hr	Model
$f_{old,r,t}$	Frequency of service in old system	Variable	1/hr	GTFS [17]
$x_{new,r,t}$	Ridership in new system	Variable	-	Model
$x_{old,r,t}$	Ridership in old system	Variable	-	Ridership [16]
$H_{block,r,t}$	Number of hours in time block	Variable	hr	Ridership [16]
$\overline{T}_{r,t}$	Average route duration	Variable	hr	GTFS [17]
$\overline{L}_{r,t}$	Average route distance	Variable	km	GTFS [17]
W_{driver}	Bus operator hourly wage rate	34.04	$/hr	City of Ott [7]
$W_{passenger}$	Passenger hourly wage rate	35.06	$/hr	JobBank [18]
$F_{wage,t}$	VTTS fraction of passenger wage	0.7-1.0	-	Various [11, 12]
P_{fuel}	Price of fuel per liter	1.3	$/liter	StatCan [8]
$P_{maintenance}$	Price of maintenance per km	0.4	$/km	StatCan [10]
FC_t	Fuel consumption rate	0.198-0.242	liter/km	NRCan [19]
$I_{GHG,car,t}$	Emission intensity of car	0.295-0.33	kgCO2e/kPT	Wang et al. [14]
$\overline{L}_{r,passenger}$	Average trip length of passenger	Variable	km	Estimate
SCC	Social cost of carbon	0.275	$/kgCO2eq	GoC [20]
$V_{km,saved}$	Vehicle km saved per 1 km of passenger transit travel	5.8	km	Newman [13]

3 Implementation

3.1 Model Structure

The optimization model was implemented in Python using the Gurobi Optimizer. Gurobi is a commercial solver that is well suited to solving MILPs. For user clarity, the implementation follows a modular architecture. Each major component is given its own class. This was helpful during iteration, particularly during sensitivity analysis where branching changes were common.

The code is organized into modular components for costs, benefits, and demand, each expressed as a linear function of the decision variable $n_{\text{new},r,t}$. The coefficients used in the benefit terms are pre-computed from the Taylor linearization described previously, ensuring that the final optimization problem is fully linear.

All model parameters (e.g., route characteristics, ridership, and cost coefficients) are stored in dictionaries indexed by (r, t), allowing efficient evaluation across the full problem space and straightforward modification for sensitivity analysis.

3.2 Model Construction

Decision variables are defined over all route–time block pairs $(r, t) \in \mathcal{R} \times \mathcal{T}$ as integer variables:

$$n_{\text{new},r,t} \in \mathbb{Z}_{\geq 0} \tag{27}$$

These variables are created using Gurobi's indexed variable interface. The objective function and constraints are then assembled programmatically using Gurobi's API, maintaining a direct correspondence with the formulation.

3.3 Solution and Post-Processing

The resulting problem is a MILP with 510 decision variables and a highly decomposable structure.

The optimized solution is then post-processed to compute changes in ridership, costs, and benefits. These results are aggregated by route and time block and visualized to support interpretation of the system-wide impacts.

4 Results

4.1 Standard-Configuration Summary

Given our standard configuration as defined by Section 2.3.8 and Table 3, the results are summarized in Table 4. The large negative objective value (relative to the budget used) indicates that the benefits of ridership far outweighed the costs. The system estimates an increased ridership of $\approx 40,000$ across an average weekday. Compared with the average daily ridership as defined in Section 2.5.1, this corresponds to a 20.0% increase.

Table 4: Summary of optimization results under standard configuration

Metric	Value	Metric	Value
Net objective value	-356,153	Total ridership increase	41,366
Total benefit change	368,884	Total cost change	12,731
Budget used	12,731	Budget slack	37,269
Max fleet slack (time block)	0	Min fleet slack (time block)	0
Peak ridership gain (Midday)	12,735	Lowest ridership gain (Early AM)	2,198
Solver node count	1	Optimality gap	0

Presuming a 1.1 expansion factor (i.e. 10% increase in bus fleet since 2023) for the fleet size for each time block, the resulting slack remains 0 across all time blocks. This indicates that the fleet constraint is binding – and even more so if the fleet has not acb tually been expanded.

Gurobi solves the model to optimality very efficiently, consistently requiring minimal branch-and-bound effort and achieving a zero optimality gap in negligible runtime.

4.2 Use of Route Subset for Analysis

A subset of four Carleton-interfacing routes (10, 111, 48, and 7) is used for visualization and discussion. While the optimization is performed over the full dataset, presenting all routes simultaneously results in figures that are difficult to interpret.

The selected routes provide a meaningful cross-section of demand patterns relevant to the project scope, especially with respect to commuting to and from Carleton. As shown in Figure 1, these routes exhibit varying baseline service levels and respond differently to optimization, allowing for clearer qualitative insights.

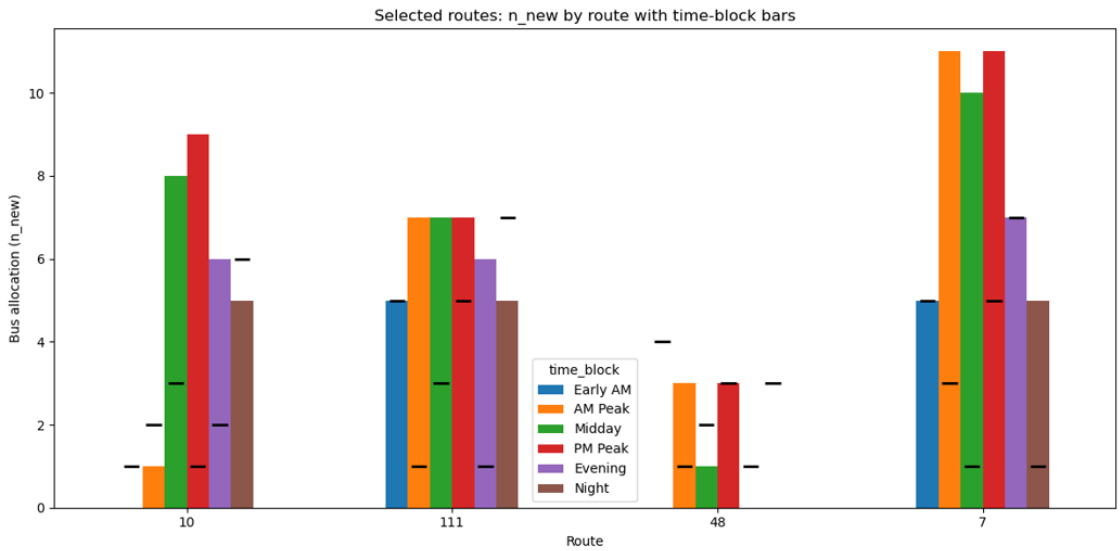

Figure 1: Key result summary for Carleton routes.

From the subset results, a consistent trend is observed in which higher-demand routes (e.g., routes 10 and 7) receive increased bus allocations, particularly during peak and midday time blocks. In contrast, lower-demand routes such as route 48 experience reduced or minimal changes in service. This reflects the structure of the objective function, which prioritizes allocations that yield the greatest marginal benefit.

4.3 Sensitivity Analysis

Sensitivity analysis was conducted to evaluate the impact of key modelling assumptions on the optimized solution.

4.3.1 Fleet Constraint Sensitivity

Figures 2 and 3 show the resulting allocations for $\Delta n_{max} = 2$ and $\Delta n_{max} = 4$, respectively, across a range of expansion factors. In both cases, the optimized number of buses assigned to each route remains largely bound by Δn_{max} as the expansion factor increases.

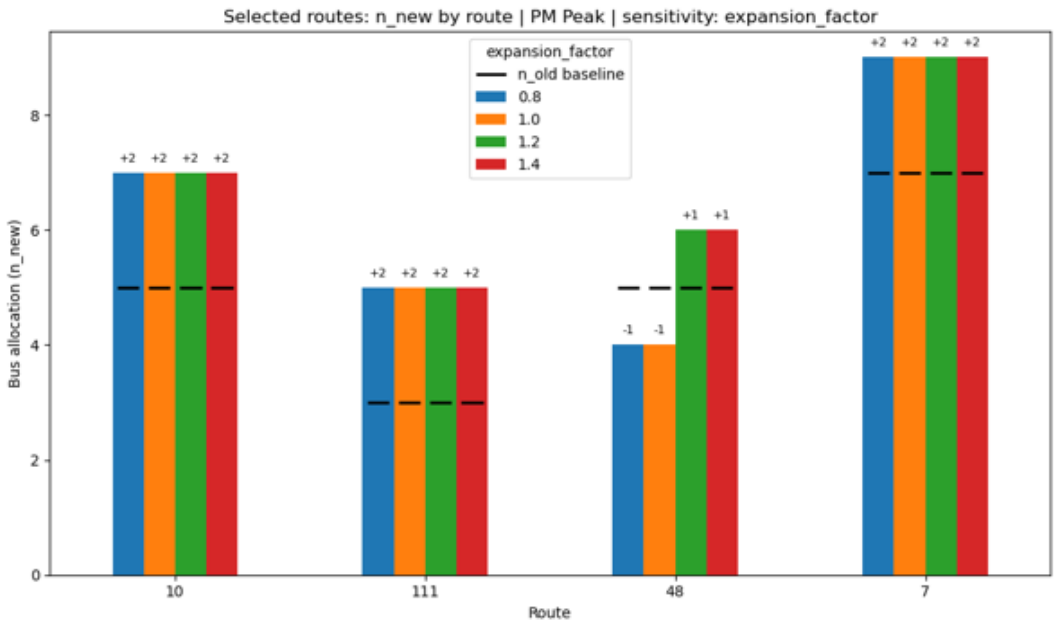

Figure 2: $\Delta n_{max} = 2$

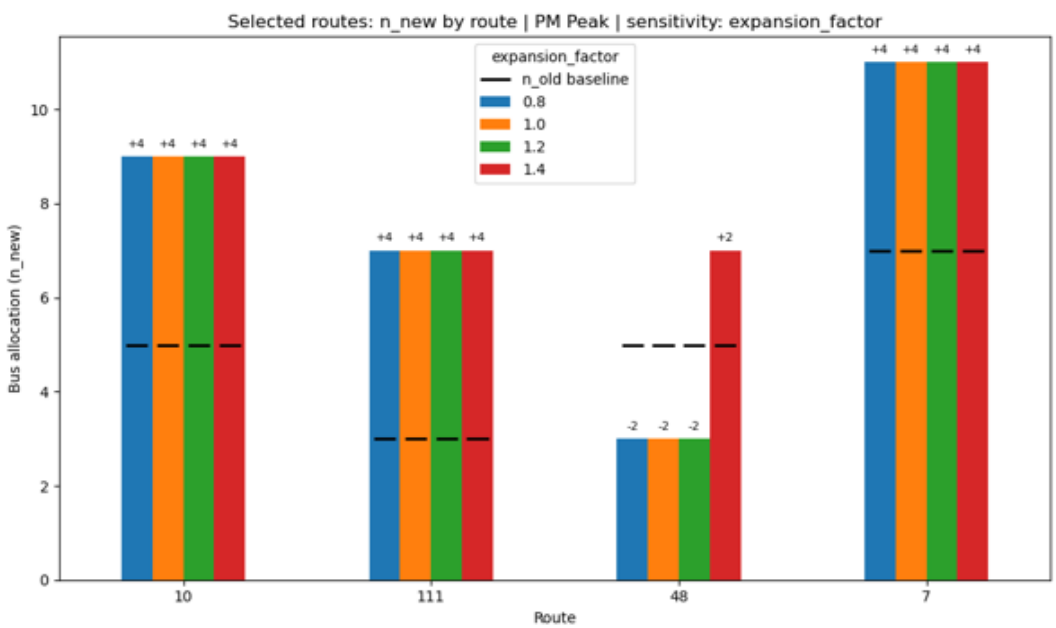

Figure 3: $\Delta n_{max} = 4$

Increasing Δn_{max} from 2 to 4 allows for larger changes in bus allocation, which is reflected in the higher absolute values of $n_{\text{new},r,t}$. However, the relative distribution of buses across routes remains consistent. Routes with higher baseline ridership (e.g., routes 10 and 7) continue to receive increased allocations, while lower-demand routes show limited response.

4.3.2 Budget Sensitivity

Sensitivity analysis revealed that the results are largely insensitive to the budget constraint as compared to the fleet constraints. As shown in Figure 4, the net change in costs must actually be a *reduction* to produce a change in the result in the Carleton subset.

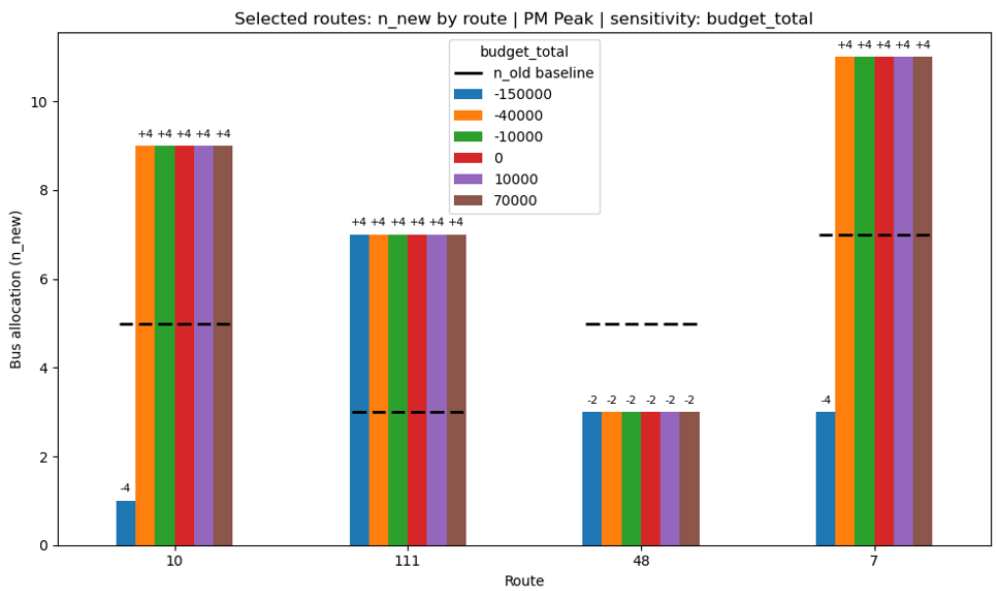

Figure 4: Sensitivity of results to total budget.

4.3.3 Cost and Benefit Parameters

Sensitivity analysis was similarly performed on the parameters in the cost and benefit equations (i.e., the constants in Table 3. The results, included in Appendix A.3, show that for the Carleton subset, the results are identical across multipliers from 0.8 - 1.4. This further suggests that Δn_{max} and the linearization assumption are the most prevalent factors in the results.

4.4 Key Findings

The solution is consistently constrained by fleet allocation limits rather than by budget or parameter estimation. Even when additional fleet is made available through the expansion factor, the optimal allocation does not change significantly. This indicates that the marginal benefit of reallocating buses is quickly saturated under the current model structure.

The optimization strongly favours high-ridership routes. Without constraints on Δn, this behaviour causes extreme reallocation, including the elimination of service for the majority of routes favour of the top performers. The linearization of the demand and benefit functions enable this issue. As the model assumes a constant marginal benefit with respect to $n_{\text{new},r,t}$, it does not capture diminishing returns associated with large increases in service frequency.

5 Conclusion

In summary, a MILP was developed to determine the optimal number of buses to assign to a given OC Transpo route based on maximizing socio-economic benefits. To achieve this, non-market valuation techniques were used to quantify benefits associated with reduced passenger wait times and avoided emissions from a mode shift from personal vehicles to transit. The costs considered include maintenance, fuel, and driver wages. Using current ridership data, future ridership was projected based on changes in service frequency, assuming a demand elasticity of 0.5. GTFS data was used to characterize the current system.

To solve the MILP, Gurobi was integrated into a Python script. Given the optimized system, a 20.0% increase in ridership is expected during weekdays. Sensitivity analysis was conducted to assess the impact of cost and constraint parameters on the solution. The results showed that the solution is primarily constrained by fleet capacity limits and restrictions on the maximum allowable changes in the number of buses assigned to each route. This constraint is particularly important, as without it the objective function would favour only high-ridership routes, potentially eliminating service on lower-ridership routes. The results further demonstrate that the conclusions are robust to a reasonable degree of uncertainty in both cost and benefit parameters.

5.1 Future Work

The most urgent limitation to resolve is the linearization of the problem, which overestimated the benefit of added buses on popular routes. In future work, this could be addressed with either a more robust linearization such as one implementing a step function. Alternatively, the problem could be reformulated as a Mixed Integer Non-linear Program (MINLP), which would eliminate the need for linearization.

Another key limitation of this project is the quality of the ridership data, which affects the accuracy and applicability of the results. Future work should prioritize obtaining more current data, ideally averaged over a longer time period rather than a single sample week in January. Additionally, access to higher-resolution data, such as hourly ridership instead of aggregated time blocks, would enable more precise optimization of the system.

In addition to this, a more complete evaluation of benefits and costs associated with a change in frequency could be added to the model equations. Such areas of refinement would include modelling the impact of lower congestion on the roads especially during peak times, creating a more complete representation of the value of time saved based on factors such as crowdedness and accounting for the fact that a portion of the modeshift represented in the problem may be to the train and not bus routes.

Finally, this model does nothing to differentiate the difference in types of buses that are part of the OC Transpo fleet. For example the fleet includes two storey buses, accordion buses and standard buses. In the current model there is no consideration of how the type of bus effects the capacity a given bus can handle.

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A Appendix

A.1 Data

Route	Wkdy Early AM	AM Peak	Midday	PM Peak	Evening	Night	**Wkdy**
1							**60600***
R2	43	1039	2231	1628	786	285	**6013**
5	23	292	566	303	162	73	**1419**
6	177	1724	3513	2781	1064	448	**9707**
7	110	1649	3533	2555	1075	550	**9472**

Figure 5: Excerp from route-level OC Transpo ridership data [16]

A.2 Complete Dataset Results

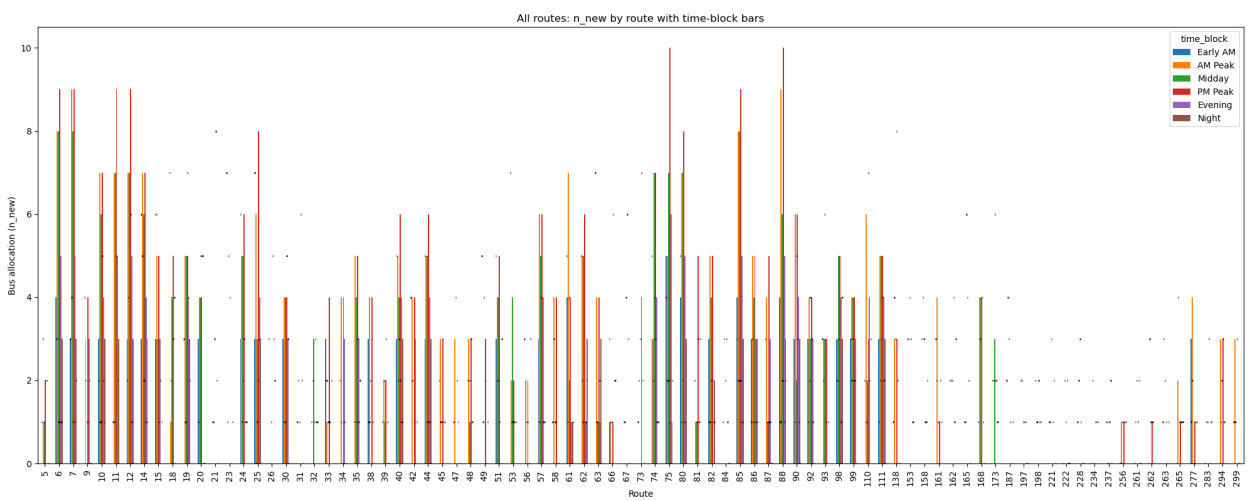

Figure 6: Standard-configuration solution displaying all routes. The least in-demand 15% have been eliminated

A.3 Stagnant Sensitivity Analysis

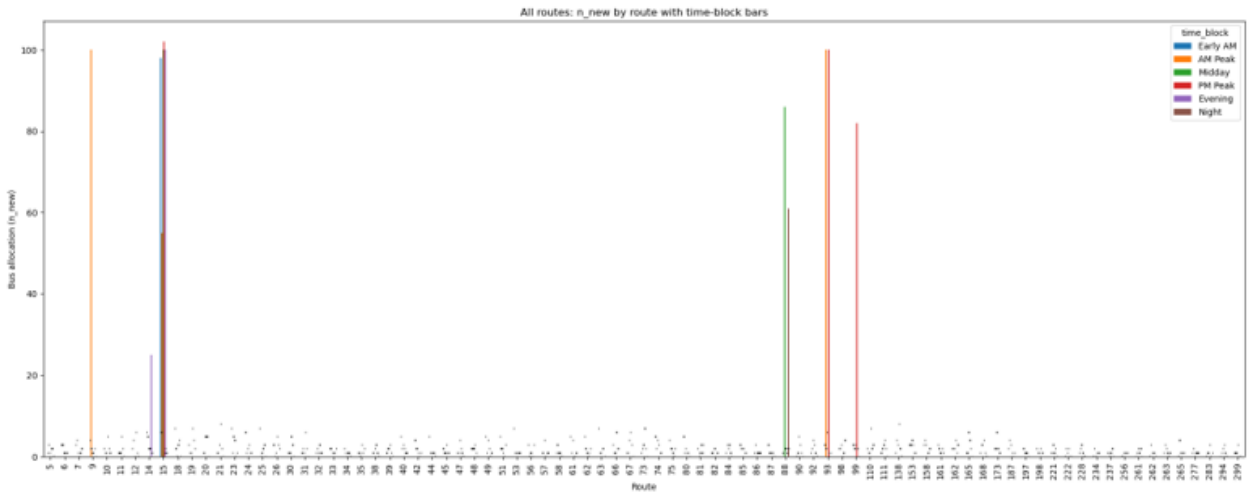

Figure 7: Unrestrained-Δn-configuration solution displaying all routes. The few non-zero bars represent the highest-demand routes, which have been rewarded with 100 buses while the other routes are eliminated entirely

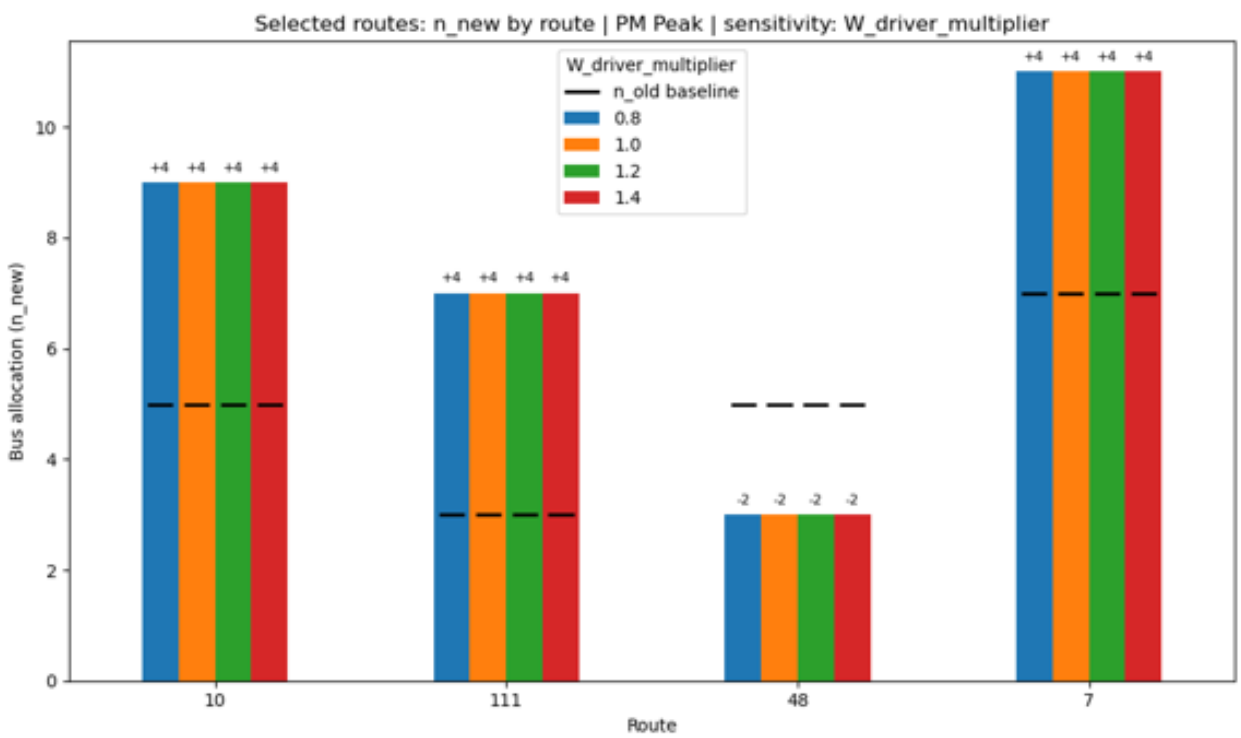

Figure 8: Sensitivity to driver wage.

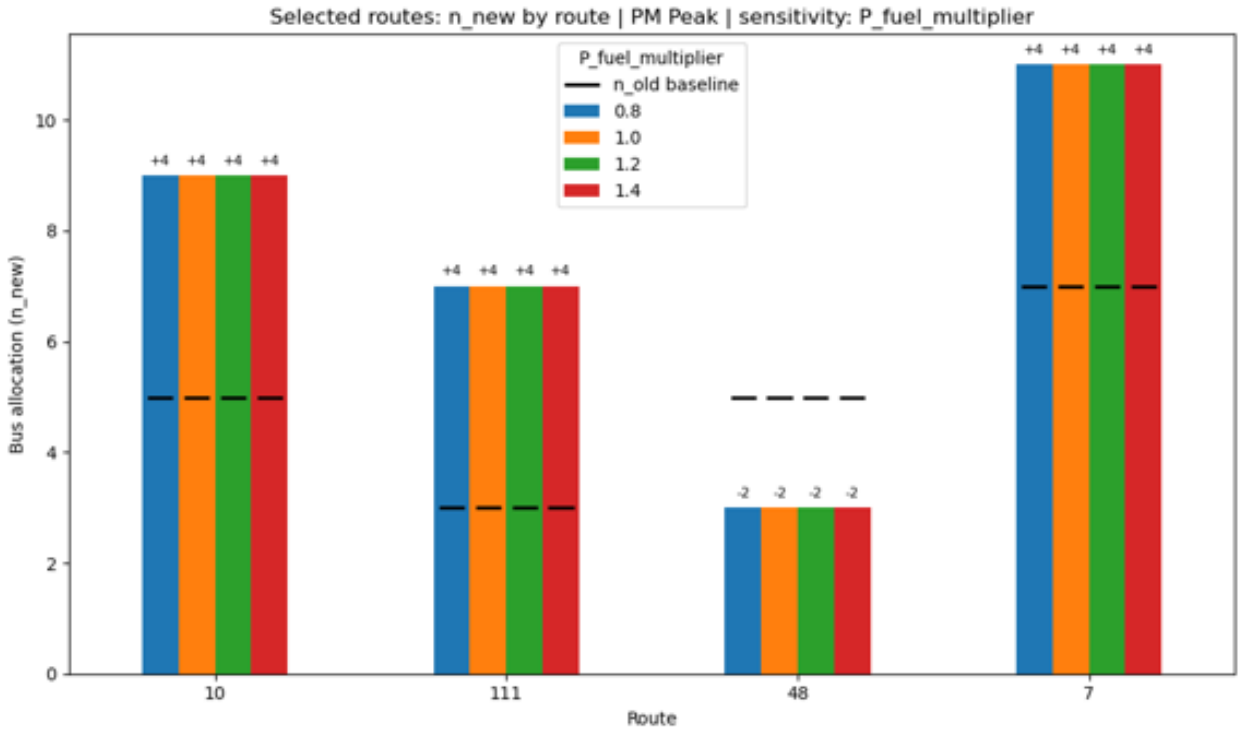

Figure 9: Sensitivity to fuel price.

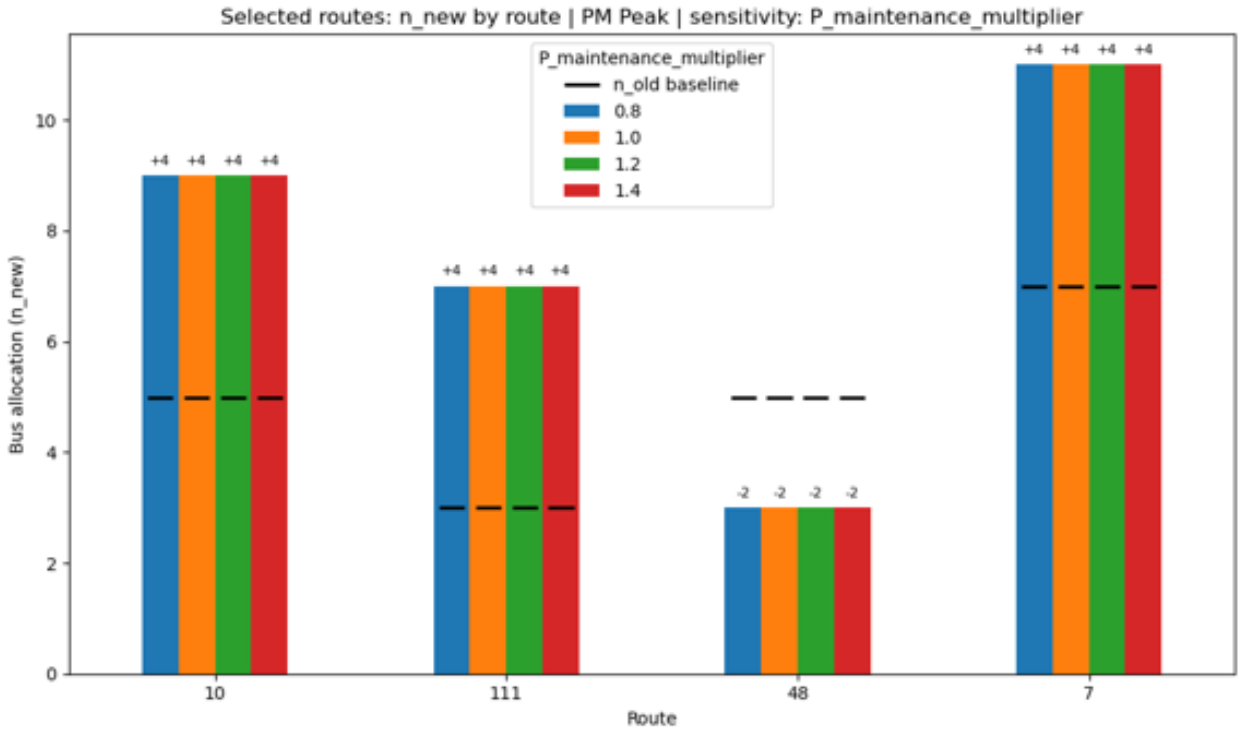

Figure 10: Sensitivity to maintenance cost.

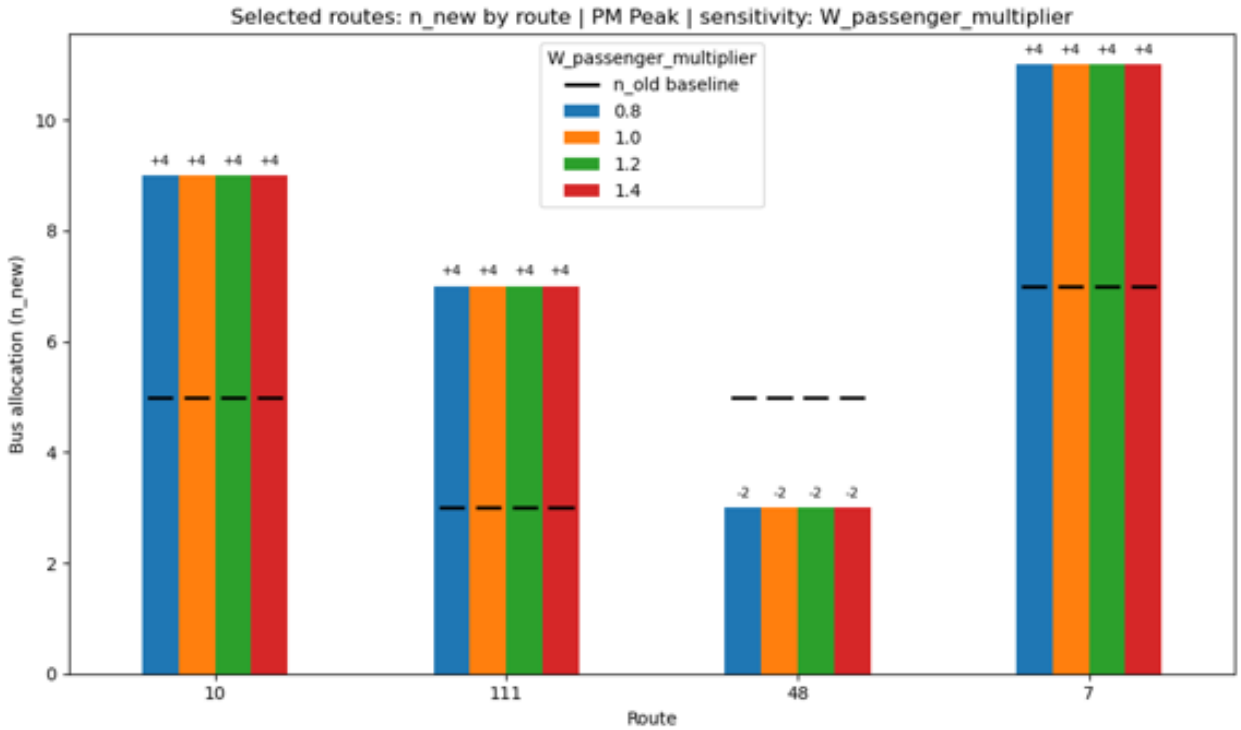

Figure 11: Sensitivity to passenger wage (value of time).

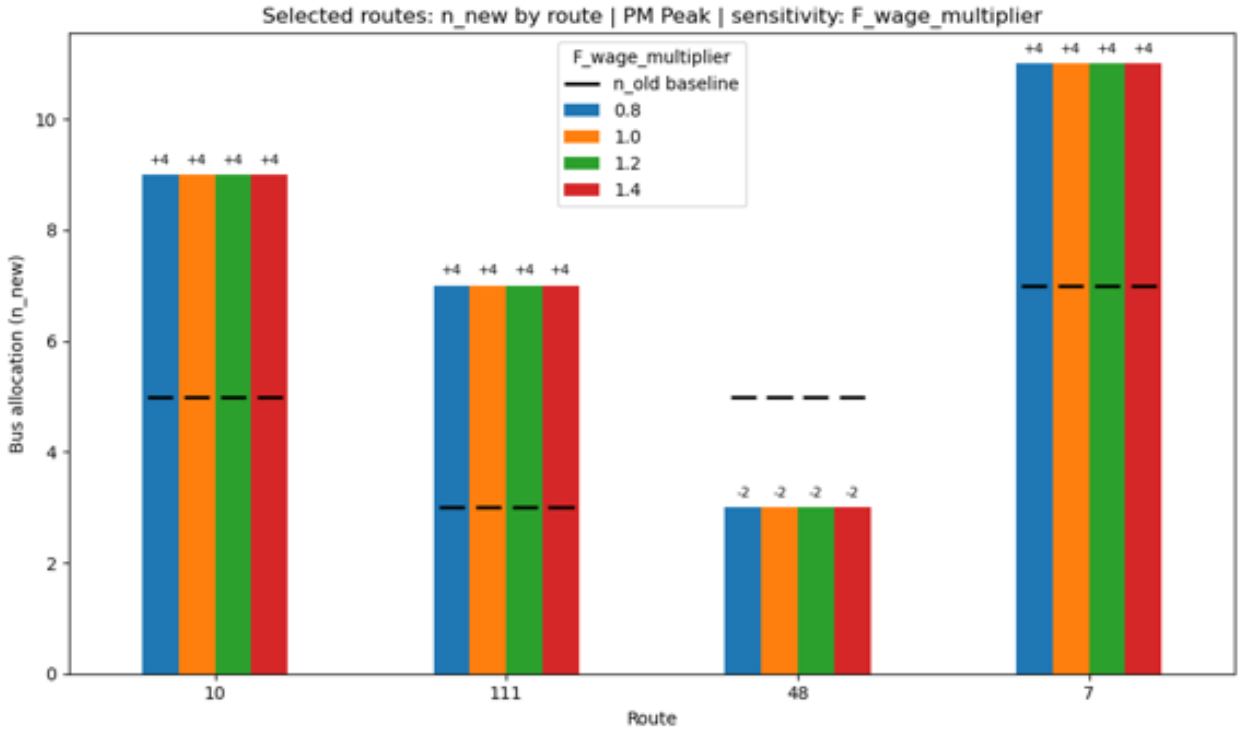

Figure 12: Sensitivity to value-of-time fraction.

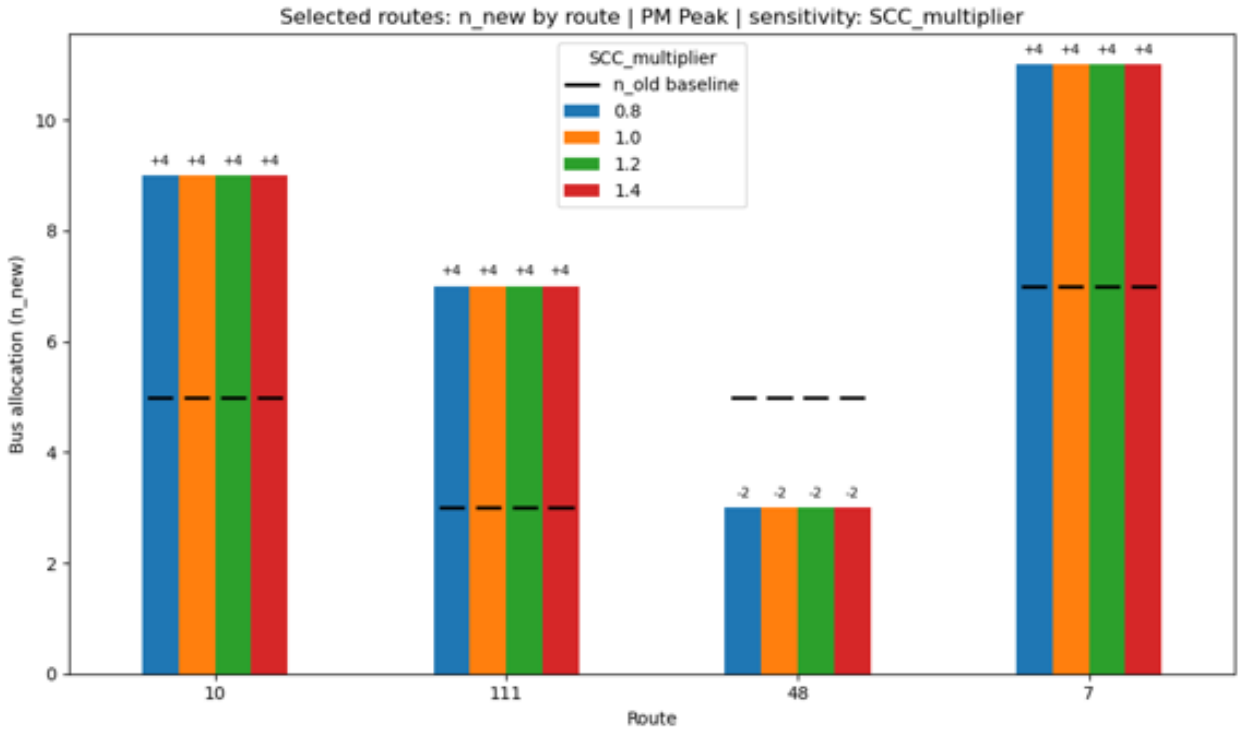

Figure 13: Sensitivity to social cost of carbon.

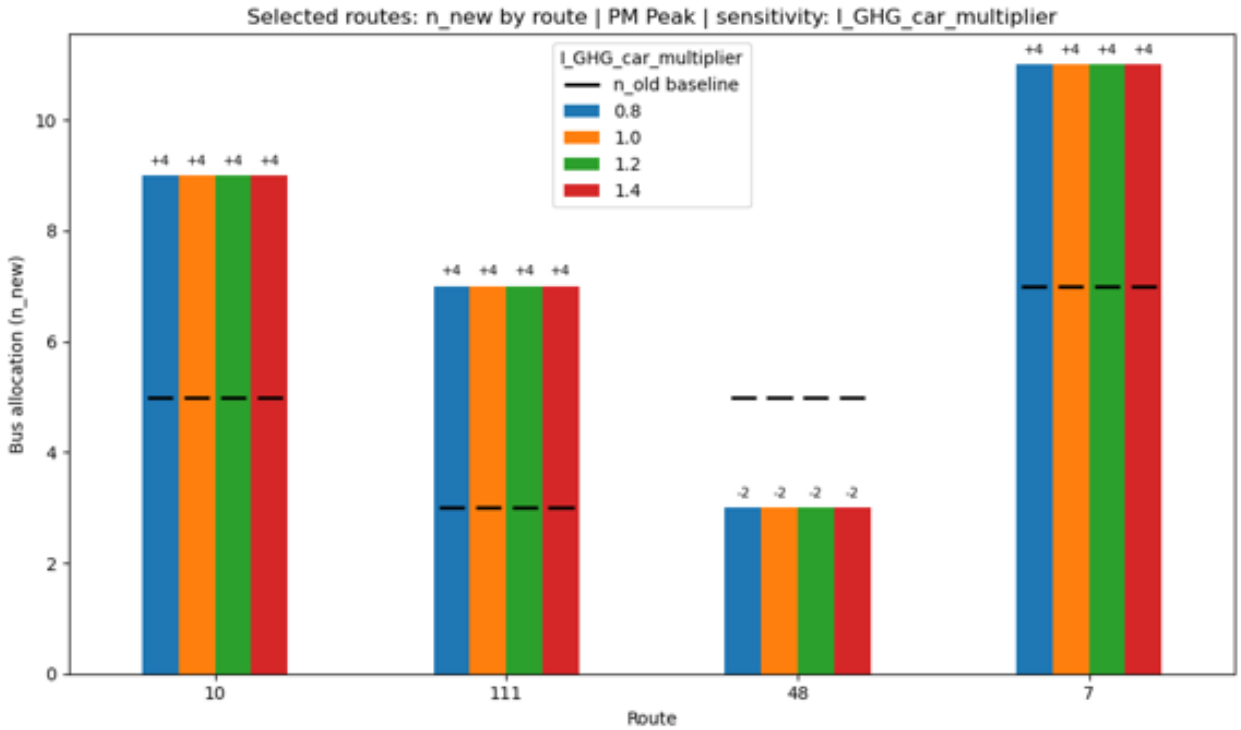

Figure 14: Sensitivity to emissions intensity.

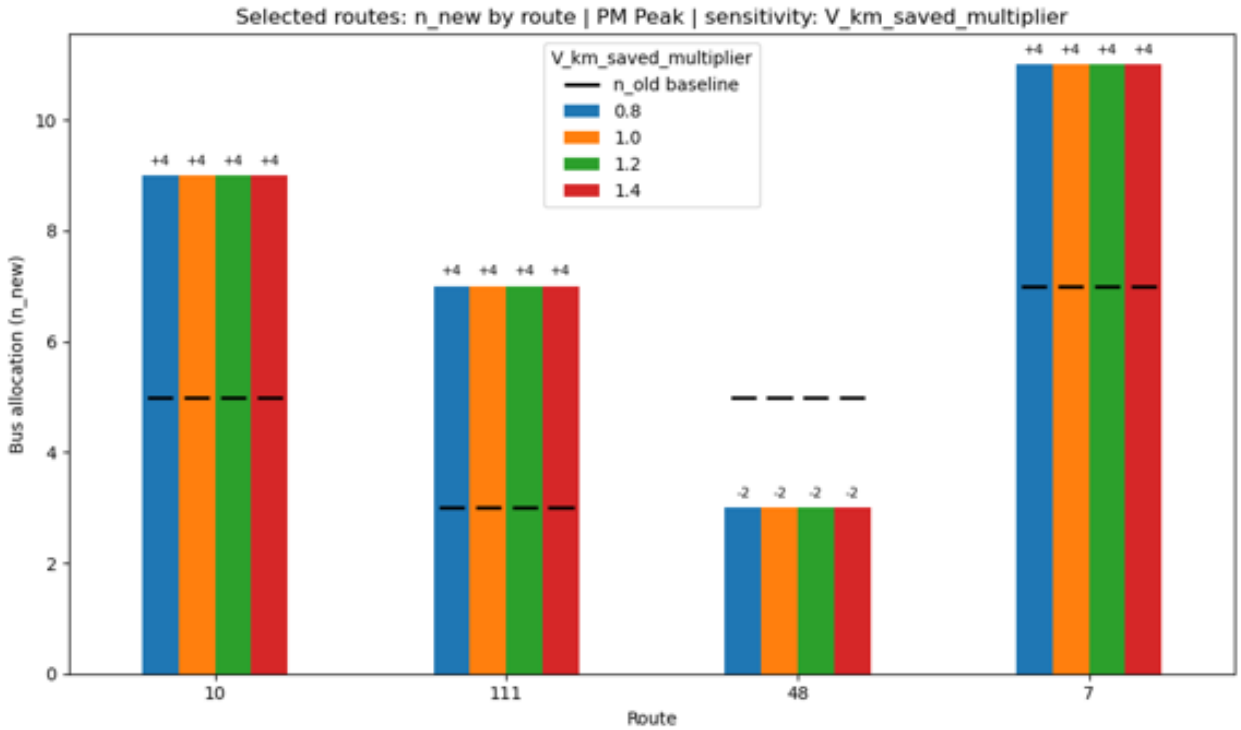

Figure 15: Sensitivity to vehicle-kilometers saved factor.

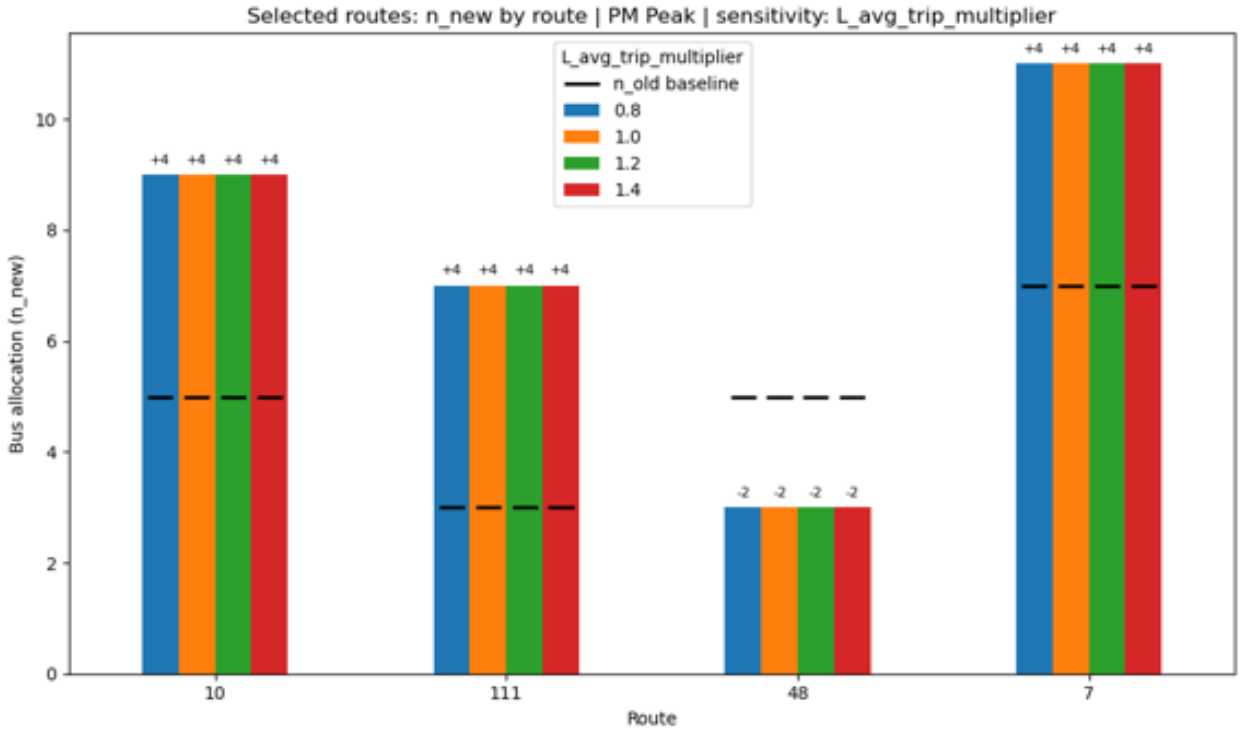

Figure 16: Sensitivity to average trip length.

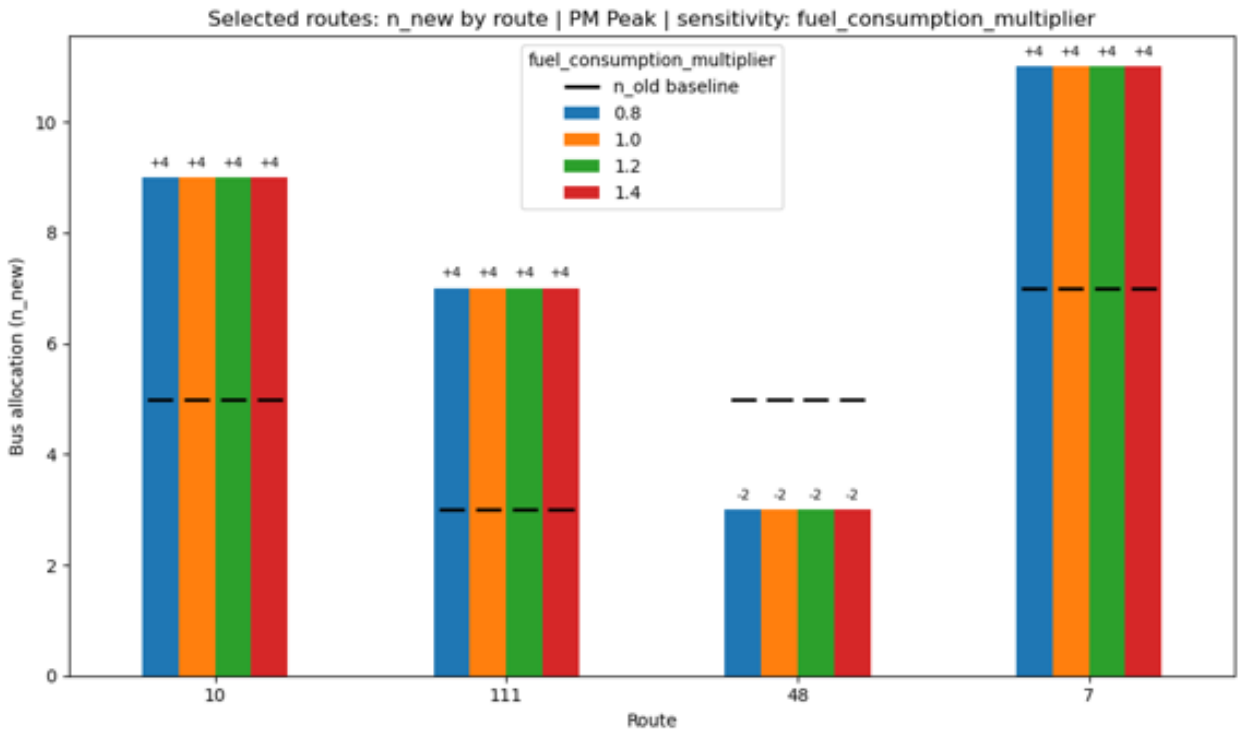

Figure 17: Sensitivity to fuel consumption.